

# Tanzim Hossain Safin

AI SECURITY & PRIVACY RESEARCHER

Dhaka, Bangladesh · [tanzimhossainsafin756@gmail.com](mailto:tanzimhossainsafin756@gmail.com) · +880 1752 329950  
[github.com/TanzimHossainSafin](https://github.com/TanzimHossainSafin) · [linkedin.com/in/tanzim-hossain-safin](https://linkedin.com/in/tanzim-hossain-safin)

## SUMMARY

Computer Science graduate from BRAC University specializing in the security and privacy of AI systems, with a focus on Retrieval-Augmented Generation (RAG) and multi-agent LLM pipelines applied to sensitive, high-stakes data. Proficient in Python, TypeScript, and Node.js, with hands-on experience designing distributed systems, running empirical attack/defense evaluations, and building scalable backend services. Passionate about applying rigorous, metrics-driven research to make AI systems trustworthy in domains like healthcare, where privacy and reliability directly affect real people.

## RESEARCH INTERESTS

RAG Systems, LLM Security & Privacy, Multi-Agent LLM Pipelines, Membership Inference & Extraction Attacks, Differential Privacy, Distributed Systems Security

## EDUCATION

**BRAC University**, Merul Badda, Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering (CSE)

Cumulative GPA: 3.63/4.00

Thesis: *A Security Evaluation Framework for Distributed Retrieval-Augmented Generation Systems*

## THESIS

**A Security Evaluation Framework for Distributed Retrieval-Augmented Generation Systems**

Advisor: Dr. Swakkhar Shatabda

- Developed a seven-step, architecture-agnostic security evaluation framework for distributed RAG systems, deriving six attack classes from their unique distributed components: Data Poisoning, Source Selection Manipulation, Knowledge Extraction, Membership Inference, Selective Forwarding, and Denial of Service.
- Evaluated the framework on two structurally different systems, DRAG and Reliable-dRAG, revealing significant vulnerabilities: knowledge extraction recovered up to 42% of documents, membership inference reached AUC-ROC 0.90, and poisoning reduced DRAG F1 by up to 24 points while achieving 100% attack success on Reliable-dRAG.
- Uncovered Grounding-Farming, a novel attack that manipulates blockchain-based reliability scores while bypassing the system's existing defense.
- Designed and independently evaluated six targeted defenses for DRAG, restoring poisoning-degraded F1 to 80–84%, reducing extraction to  $\leq 0.02\%$ , lowering membership-inference AUC-ROC from 0.88 to 0.76, and reducing dropped queries by up to 74.6%.
- Swept defenses across attack intensities, finding that three defenses became stronger as attacks intensified, while the congestion-based DoS defense weakened under heavy attack load.

## RESEARCH & PROJECTS

**Trust Propagation and Structural Containment in Multi-Agent LLM Pipelines**

*Accepted, IEEE CARS 2026*

- Problem:** In multi-agent LLM systems, a compromised low-privilege agent can pass forged authority down the chain and cause a high-privilege agent to take a real, unauthorized action.
- Solution:** Built a four-agent LangGraph pipeline (Supervisor, Researcher, Validator, Executor) and evaluated two real attacks memory poisoning and indirect prompt injection against both the Validator's judgment and an independent, cryptographically-signed authorization layer.
- Result:** The structural authorization layer cut unsafe execution from 100% to 0% across both attacks without harming legitimate task completion. Introduced the Judgment Bypass Rate, a new metric that separates a fooled reviewer from an actual downstream safety failure.

### AI-Powered Financial System

- Problem:** Individuals managing personal finances often lack a single, intelligent tool to track spending, assess risk, and plan investments, leading to fragmented budgets and poor financial literacy.
- Solution:** Built a full-stack AI-powered financial platform (React/Tailwind frontend, Node.js/Express backend) for budgeting, investment tracking, and expense optimization, using intelligent recommendations to help users understand spending habits and plan investments.
- Features:** Automated budget organization, spending-habit analysis, investment planning, and risk assessment, with AI-driven recommendations surfaced through a REST API and a React-based interface.

## PUBLICATIONS

- Tanzim Hossain Safin, et al. **"Trust Propagation and Structural Containment in Multi-Agent LLM Pipelines."** IEEE CARS 2026. (*Accepted*)

## SKILLS

- Languages:** Python, JavaScript, TypeScript, C, C++
- AI / ML:** RAG Systems, LLM Security & Privacy, Multi-Agent Systems, Membership Inference & Extraction Attacks, Differential Privacy, Sentence-BERT Embeddings, Scikit-learn, Pandas, NumPy
- Databases:** MongoDB, PostgreSQL, MySQL
- Frameworks & Libraries:** Node.js, Express, LangGraph, React, Next.js